IIDS67692 Computational Methods for Multimodal Data Analysis

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Part B (word counts: 483)

Dataset and Preprocessing

The PathVQA dataset for Medical Visual Question Answering (VQA) consists of pathology images and question-answer pairs. After removing unused and duplicate samples, the final dataset comprises 4,289 images and 32,632 QA pairs (precise distribution in Table 1). For data preparation, images are resized to 224x224 pixels using BICUBIC interpolation to preserve critical pathological details, followed by normalization. Due to hardware constraints, a random subset of 3,000 samples was partitioned into an 80/20 training/validation split and loaded into PyTorch DataLoaders (batch size = 32), with shuffling enabled exclusively for training to prevent overfitting.

Table 1. Statistical distribution of the PathVQA dataset splits.

	Training Set	**Validation Set**	**Test Set**
QAs	19,654	6,259	6,719
Images	2,599	832	858

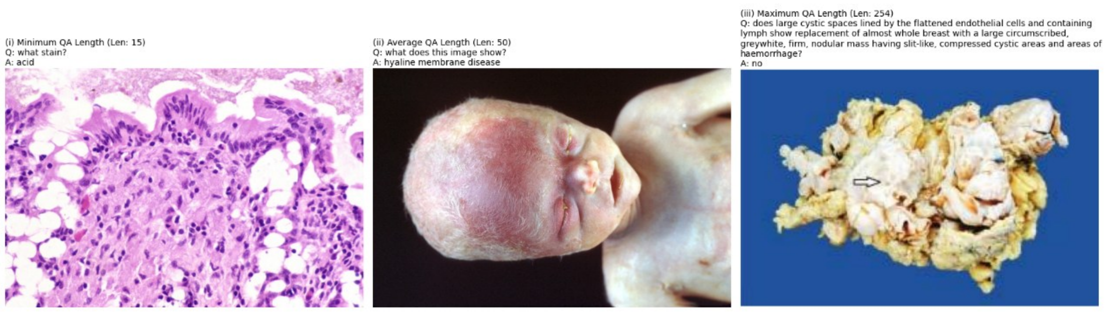

Figure 1. PathVQA Dataset Examples Exhibiting Selected QA Length Milestones. Illustrations depict individual cases for (i) Minimum QA length (15 tokens/characters; Index: 2093), (ii) Dataset average QA length (50 tokens/characters; Index: 282), and (iii) Maximum QA length (254 tokens/characters; Index: 6132), showing the paired medical imagery and QA text annotations.

Methodology

Four fusion approaches were evaluated within a GPT-2 decoder framework: Cross-Attention (CA), Gated Cross-Attention (Gated CA), Parallel Cross-Attention, and CLIP. The **Gated Cross-Attention** design extends standard CA by introducing a learned sigmoid gate applied element-wise to the cross-attention output before the residual addition. This gate modulates how much visual information is injected at each transformer block, selectively suppressing irrelevant image features, a mechanism especially relevant in medical imaging where questions may concern only localised pathological regions.

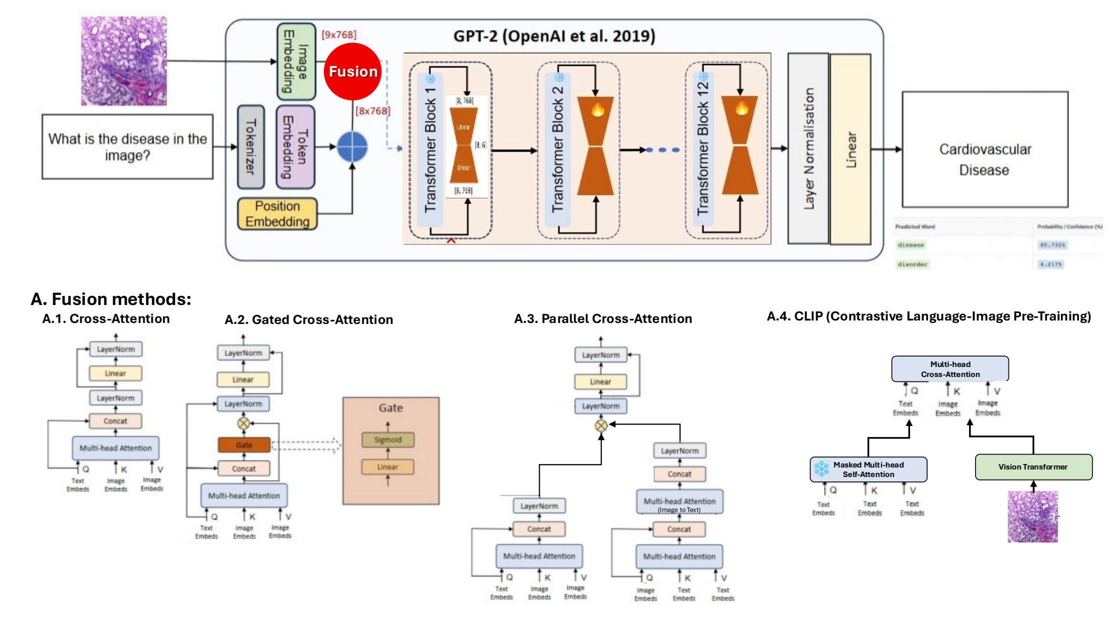

Figure 2. Model architecture for different fusion approaches.
The framework integrates an image encoder and text tokenizer with a GPT-2 language model decoder. Sub-panels illustrate four distinct feature fusion layers investigated during hyperparameter tuning: (A.1) Cross-Attention, (A.2) Gated Cross-Attention, (A.3) Parallel Cross-Attention, and (A.4) the CLIP (Contrastive Language–Image Pre-training) encoding configuration.

Results

Group A.

CA achieves the strongest qualitative accuracy with fully correct predictions (BLEU-1 = 0.3835, Rouge-L = 0.4730). Gated CA produces semantically adjacent but incorrect outputs (e.g., "heart" instead of "cardiovascular"), as the additional gate parameters require more data or training steps to converge, yielding slightly lower scores (BLEU-1 = 0.2114). Parallel CA overfits severely (BLEU-1 = 0.1112), while CLIP achieves competitive metrics but exhibits factual errors, indicating its contrastive pretraining objective misaligns with open-ended generative medical QA.

Group B.

Using Gated CA as baseline, three hyperparameter axes were explored. For **learning rate** (B-1), the default lr = 2×10^{-4} represents the optimal balance; a higher rate (lr = 2×10^{-3}) produces deceptively low loss (1.7532) but entirely incorrect predictions due to unstable weight updates that prevent meaningful cross-modal alignment, while an excessively low rate (lr = 2×10^{-5}) causes underfitting (loss = 3.4169) as the model fails to sufficiently update the LoRA adapters within the available epochs. For **LoRA rank** (B-2), reducing rank to 4 improves prediction correctness over the default rank 8, because a smaller parameter subspace forces the model to learn only the most task-relevant low-rank adaptations and discards noise-inducing redundancies; rank 16 introduces sufficient over-parameterisation to incur one prediction error despite higher capacity. For **attention heads** (B-3), scaling to 8 heads yields the highest Rouge-1 (0.5145) and Meteor (0.2610), as finer-grained parallel attention sub-spaces improve syntactic fluency; however, both 2 and 8 heads still incur one qualitative error, demonstrating that head count primarily governs output fluency rather than cross-modal factual grounding.

Conclusion

Quantitative metrics and qualitative samples must be interpreted jointly in medical VQA. The optimised Gated CA (8 attention heads) surpasses CA and CLIP on Rouge-1 and Meteor, demonstrating that the gating mechanism realises its advantage only after targeted tuning, particularly smaller LoRA rank and appropriate learning rate, which together stabilise cross-modal feature integration and constrain unnecessary parameter growth.

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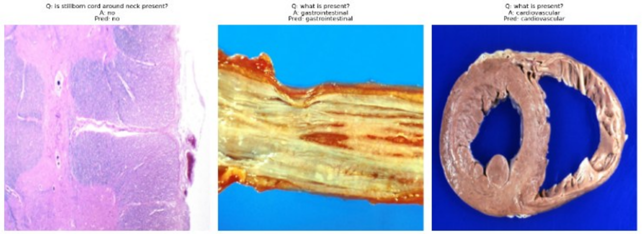
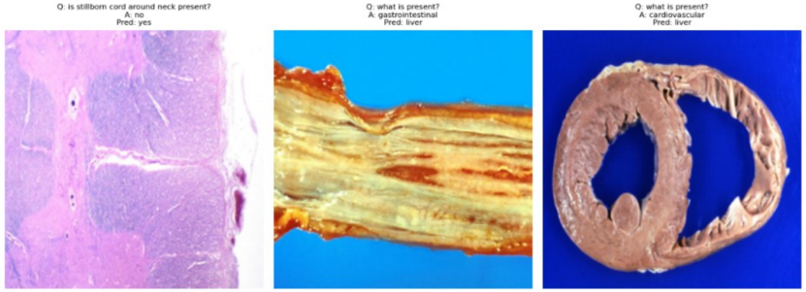
<table>
<tr><th colspan="2">(a) Comparison of different multi-modal fusion methods (Group A)</th><th colspan="2">(b) Impact of different learning rates (Group B1)</th></tr>
<tr><td>(A.1)
Cross-Attention</td><td>
</td><td rowspan="2">Lr= 2x10^-3</td><td rowspan="2">
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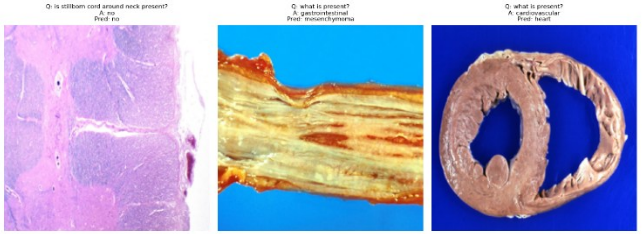
<tr><td>(A.2) Gated
Cross-Attention</td><td>
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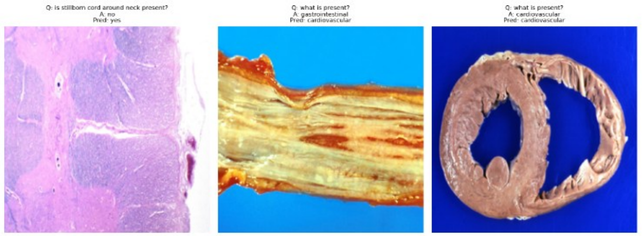
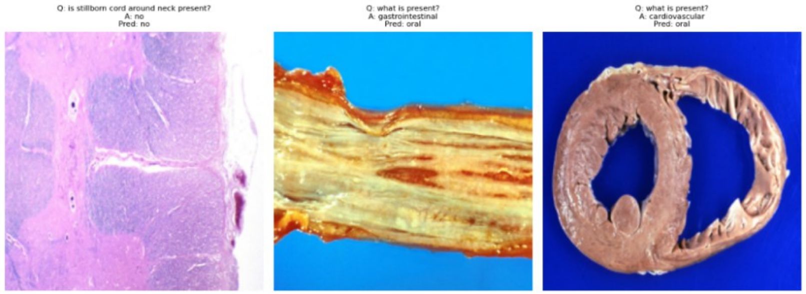
<tr><td>(A.3) Parallel
Cross-Attention</td><td>
</td><td rowspan="2">Lr= 2x10^-5</td><td rowspan="2">
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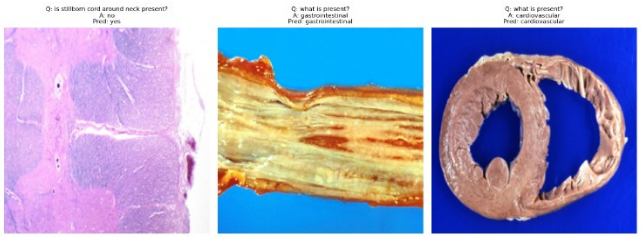
<tr><td>(A.4) CLIP</td><td>
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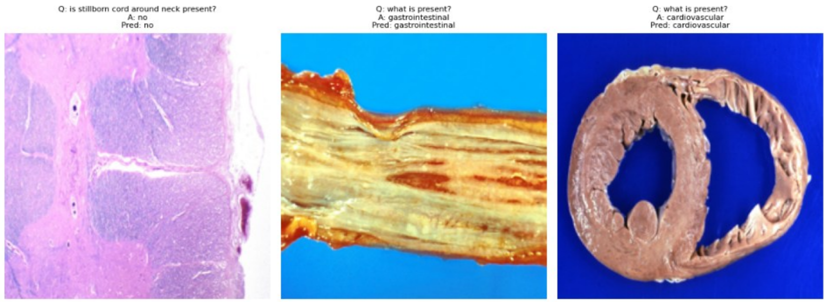
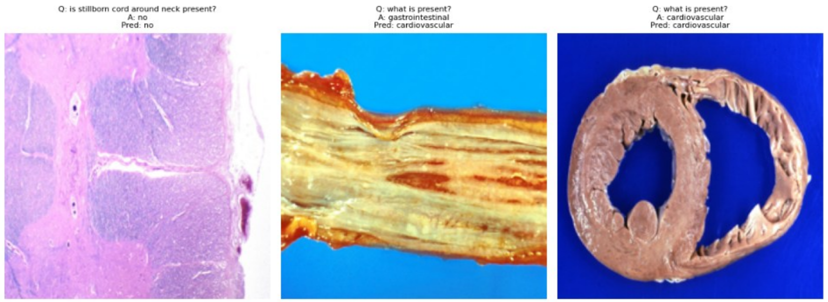
<tr><th colspan="2">(c) Effects of varying LoRA ranks (Group B2)</th><th colspan="2">(d) Influence of the number of cross-attention heads (Group B3)</th></tr>
<tr><td>LoRA = 4</td><td>
</td><td>number of attention heads = 2</td><td>
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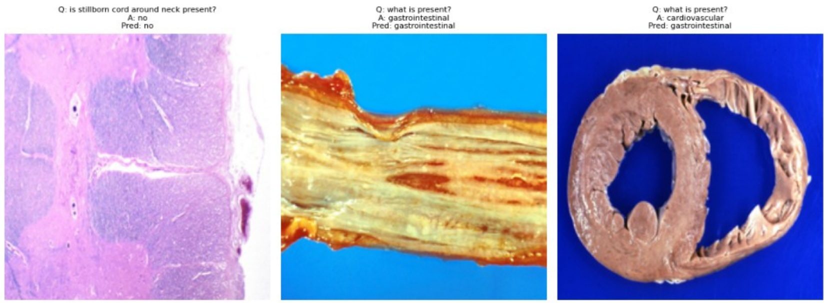
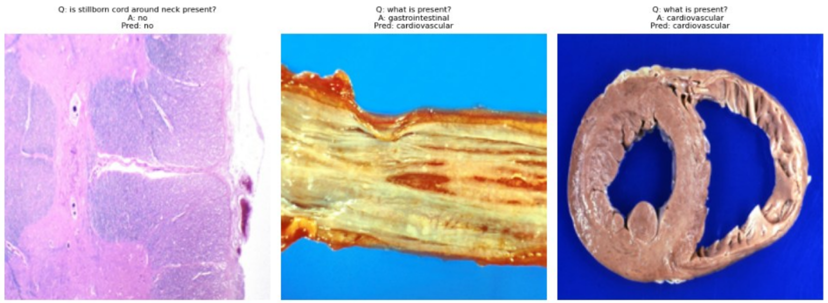
<tr><td>LoRA = 16</td><td>
</td><td>number of attention heads = 8</td><td>
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Figure 3. Qualitative comparison of multi-modal fusion methods and hyperparameter configurations.
(a) Comparison of different multi-modal fusion methods (Group A): Visual results of three selected samples across four configurations: (A.1) Cross-Attention yields fully correct predictions, demonstrating high accuracy; (A.2) Gated Cross-Attention produces an incorrect yet semantically relevant prediction (e.g., predicting "heart" instead of "cardiovascular"); (A.3) Parallel Cross-Attention and (A.4) CLIP exhibit sub-optimal performance with notable errors. **(b) Impact of different learning rates (Group B1):** Visual comparison of model predictions under varying learning rates, where a lower rate delivers superior prediction accuracy compared to a higher rate. **(c) Effects of varying LoRA ranks (Group B2):** Visual evaluation of model performance across different LoRA ranks, demonstrating that a smaller rank size contributes to improved prediction outcomes. **(d) Influence of the number of cross-attention heads (Group B3):** Visual representation of predictions across different head counts, illustrating that variations in attention heads yield no significant difference in performance.

Table 2. Quantitative text generation evaluation and loss performance across varying multimodal fusion mechanisms and hyperparameter configurations. BLEU, Rouge-1, Rouge-L, and Meteor are standard evaluation metrics for text generation quality, where higher values indicate better performance. Model 2 (Gated Cross-Attention) serves as the control baseline for all hyperparameter tuning sub-experiments in Group B.

		Best Average Training Loss	BLEU-1	BLEU-2	Rouge1	RougeL	Meteor
Group A: Architectural Fusion Variants							
1	Cross-attention	2.2202	0.3835	0.0284	0.4749	0.4730	0.2393
2	Gated Cross-Attention	1.7645	0.2114	0.0040	0.4361	0.4345	0.2207
3	Parallel Cross-Attention	1.6369	0.1112	0.0029	0.3229	0.3238	0.1664
4	CLIP	2.1770	0.3793	0.0273	0.4949	0.4931	0.2516
Group B: Hyperparameter Tuning (Baseline: Gated Cross-Attention)							
B-1. Learning rate: default = 2x10^-4 (lr=0.0002).							
5	2x10^-3 (lr=0.002)	1.7532	0.3221	0.0127	0.4641	0.4656	0.2341
6	2x10^-5 (lr=0.00002)	3.4169	0.2878	0.0000	0.2939	0.2933	0.1469
B-2. LoRA: default = 8							
7	LoRA = 4	2.4327	0.3151	0.0134	0.4472	0.4467	0.2253
8	LoRA = 16	2.0155	0.3133	0.0158	0.4388	0.4366	0.2218
B-3. cross-attention heads: default = 4							
9	num_head fussion = 2	1.9781	0.3989	0.0207	0.4699	0.4679	0.2360
10	num_head fussion = 8	2.2147	0.3538	0.0203	0.5145	0.5150	0.2610

Reference

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